# DATA ANALYSIS PROJECT PROPOSAL

Graduation Project - DEPI Data Analysis Track

**MANUFACTURING DOWNTIME ANALYSIS**

## Project Overview

| | | |
|---|---|---|
| TEAM SIZE<br>**5 members** | DATASET<br>**Manufacturing Line Productivity** | BUSINESS REQUIREMENTS<br>**Dashboard KPIs** |
| MAIN TOOL<br>**Power BI** | FINAL DEADLINE<br>**4 Dec 2026** | FINAL DEMO<br>**Demo video included** |

## Project Goal and Tools

The goal of this project is to use **Power BI** to clean, analyze and visualize the production-line data of a manufacturing plant. We want to show how much time the line loses to downtime, which causes, operators and products are involved, and how many batches the line can complete on the next day of operation.

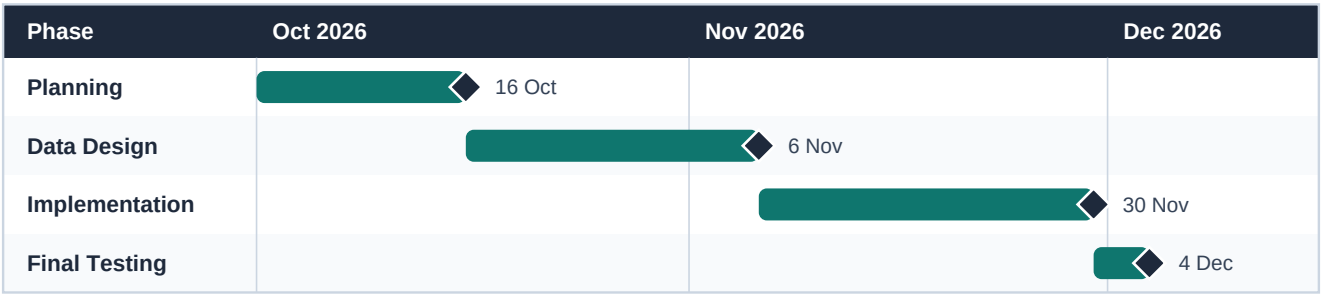
### Tools we will use

- Power BI Desktop (data model, visuals and dashboard)
- Power Query (data cleaning and preparation)
- DAX (measures and KPIs)
- Python (forecasting and prediction)
- GitHub (all project documents and files)

## Project Plan and Deadlines

| Phase | Focus | Key deliverables | Deadline |
|---|---|---|---|
| **1. Planning** | Scope, plan, roles and requirements | Project proposal, project plan, roles, risk plan, KPIs, requirements document | 16 Oct 2026 |
| **2. Data Design** | Analysis and data model design | Data model (ERD), data dictionary, analysis questions, use cases | 6 Nov 2026 |
| **3. Implementation** | Build the solution | Cleaned dataset, Power BI model and DAX measures, Python forecast, dashboard | 30 Nov 2026 |
| **4. Final Testing** | Test, report and present | Test results, final report, presentation, demo video | 4 Dec 2026 |

### Timeline (Gantt chart)




Each bar runs from the previous milestone to the phase deadline; diamonds mark the milestones. Planning starts on 1 Oct 2026.

### Milestones

| Milestone | Date | What is ready |
|---|---|---|
| **M1 Planning and requirements** | 16 Oct 2026 | Proposal, plan, roles, risk plan, KPIs, requirements |
| **M2 Data design** | 6 Nov 2026 | Data model (ERD), data dictionary and analysis questions |
| **M3 Dashboard** | 30 Nov 2026 | Cleaned data, Power BI model, DAX, Python forecast, dashboard |
| **M4 Final delivery** | 4 Dec 2026 | Test results, final report, presentation and demo video |

## Team Members and Responsibilities

| Member | Role | Responsibilities |
|---|---|---|
| **Fares Wael** | Team Lead | • Manage the project and deadlines<br>• Create and organize the GitHub repository<br>• Write the project proposal and risk plan<br>• Track tasks<br>• Clean and prepare the dataset using Power Query<br>• Create all DAX measures<br>• Check that all files are uploaded |
| **Mohamed Hazem** | Data Model and Dictionary | • Prepare the data schema/model (ERD)<br>• Define core data analysis questions<br>• Create the data dictionary<br>• Explain tables, columns, data types and relationships |
| **Mohamed Ibrahim** | Modeling, Validation and Presentation | • Data modeling and determining data analysis questions<br>• Check that model relationships work correctly across multiple sheets<br>• Help validate cleaned data<br>• Prepare the final presentation slides and present main findings and recommendations |
| **Rabab Ismail** | Forecasting and Predictive Analytics | • Forecasting logic and predictive analytics development (Python)<br>• Check calculations with the team |
| **Bassant Mohamed** | Dashboard and Report | • Plan the dashboard layout and visual design<br>• Build the dashboard pages in Power BI<br>• Write and organize the final operational report and forecasting<br>• Check that the dashboard is easy to read |

| Shared tasks (everyone) |
|---|
| • Everyone reviews the 40 questions and confirms that dataset tables have the needed columns. |
| • Everyone tests the dashboard and checks that batch totals match cleaned operational data. |
| • The team lead checks progress, but each member is responsible for finishing their own tasks. |

## Risk Assessment and Mitigation

| Risk | Impact | Mitigation |
|---|---|---|
| **Small dataset (38 batches, 4 days) makes the forecast less reliable** | High | Use a transparent method, show a best and worst case range, and label the forecast as indicative |
| **Data errors (wrong dates, duplicates, batches that cross midnight)** | High | Document every cleaning step in Power Query and run validation checks on the cleaned tables |
| **Relationships between sheets are set up wrongly** | High | Connect the tables by Batch and check that totals match with and without the relationship |
| **Different members use different definitions (efficiency, downtime %)** | Medium | Agree one definition per metric and record it in the data dictionary |
| **DAX or Python calculations give wrong numbers** | High | Compare results with the cleaned data, and have a second member review each calculation |
| **The dashboard is hard to read** | Medium | Keep one layout and colour scheme, and test it with the team before the demo |
| **Deadlines slip or files are missing on GitHub** | Medium | Track tasks weekly against the phase deadlines and check the repository before each deadline |

## Project KPIs (Measures of Project Success)

These KPIs measure the success of the project itself, not the results of the plant. The dashboard KPIs are listed later in this document.

| Project KPI | Target | How we check it |
|---|---|---|
| **Business questions answered in the dashboard** | 40 of 40, plus the next-day forecast | Tick each question in the requirements document |
| **Dashboard numbers matching the cleaned data** | 100% (38 batches, 61 downtime records) | Compare dashboard cards with the cleaned tables |
| **DAX measures checked by another team member** | All measures | A second member reviews each measure |
| **Forecast tested on past production days** | Back-test on the 4 days, with a best and worst range | Compare predicted and actual downtime (Python) |
| **Dashboard pages tested (slicers, navigation, interactions)** | All pages | Test the Date, Product and Operator slicers on every page |
| **Deliverables uploaded to GitHub** | All, before 4 Dec 2026 | Team lead checks the repository before each deadline |

| Project KPI | Target | How we check it |
|---|---|---|
| **Phase deadlines met (16 Oct, 6 Nov, 30 Nov, 4 Dec)** | 4 of 4 | Weekly progress check by the team lead |

## Dataset Overview

The workbook has two main tables. **Line productivity** holds one row per batch (date, product, operator, start and end time). **Line downtime** holds the downtime minutes of each batch by cause (12 possible downtime factors). Two small lookup sheets give the minimum batch time per product and the description of each downtime factor, including whether it is an operator error.

### Dataset at a glance

| | | |
|---|---|---|
| **38**<br>Batches (Line productivity rows) | **61**<br>Batch-by-cause records (Line downtime) | **29 Aug - 2 Sep 2024**<br>Date range |
| **4**<br>Production days | **6**<br>Unique products | **4**<br>Unique operators |

### Main source columns

| Source column | Sheet | Data type | Description |
|---|---|---|---|
| **Date** | Line productivity | Date | Production day of the batch |
| **Product** | Line productivity | Text | Product code of the batch |
| **Batch** | Line productivity | Whole number | Unique batch ID |
| **Operator** | Line productivity | Text | Operator who ran the batch |
| **Start Time / End Time** | Line productivity | Time | Clock times when the batch started and ended |
| **Factor 1 - 12** | Line downtime | Whole number (min) | Downtime minutes of the batch for each cause |

Minimum batch time comes from the Products sheet; the cause description and the operator-error flag come from the Downtime factors sheet.

## How We Will Build the Project

| | | |
|---|---|---|
| 1 | **Prepare data** | Clean and validate the raw sheets in Power Query and produce one tidy dataset. |
| 2 | **Build data model** | Load the cleaned tables and connect them by Batch with a one-to-many relationship. |
| 3 | **Create DAX** | Write the measures for production, downtime, efficiency, causes and operators. |
| 4 | **Build dashboard** | Design the pages, add the KPIs and charts, and add the Python forecast. |
| 5 | **Test and finish** | Check totals against the cleaned data, write the report and record the demo. |

### Suggested dashboard KPIs

| Suggested KPI | Definition | Visual |
|---|---|---|
| **Total Batches** | Number of batches produced | Card |
| **Downtime %** | Total downtime / total production minutes | Gauge |
| **Overall Efficiency** | Total minimum batch minutes / total production minutes | Gauge |
| **Avg Downtime per Batch** | Total downtime / number of batches | Card |
| **Operator Error Share** | Operator-error minutes / total downtime | Donut chart |
| **Top Downtime Cause** | Cause with the most downtime minutes | Bar chart |
| **Expected Batches Next Day** | Forecast from the operating minutes available (Python) | Card with slicer |

### Project checklist

| | |
|---|---|
| [ ] Clean the dataset in Power Query | [ ] Build the dashboard pages |
| [ ] Document the preprocessing steps | [ ] Validate batch totals against the cleaned data |
| [ ] Build the data model and relationships | [ ] Upload all documents to GitHub |
| [ ] Create the DAX measures | [ ] Record the demo video |
| [ ] Build the forecast in Python | [ ] Prepare the final report and presentation |